

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code: 3371	Course Title: Introduction to IC Fabrication
Semester: 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the fundamentals of IC fabrication process.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge in atomic structure & chemical Bonding	1004	Applied Chemistry	1
Basics of semiconductor physics	2003	Applied Physics II	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Summarize the basis of crystal growth, wafer preparation and wafer cleaning.	10	Understanding
CO2	Explain the process of epitaxy and oxidation.	11	Understanding
CO3	Explain lithography and etching process.	10	Understanding
CO4	Illustrate the process of deposition, diffusion, ion implantation and packaging.	12	Understanding
	Series Exam	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the basics of crystal growth, wafer preparation and wafer cleaning		
M1.01	Explain the concept of clean room in IC fabrication	2	Understanding
M1.02	Explain the methods of single crystal growth	3	Understanding
M1.03	Explain chemical-mechanical polishing technique	2	Understanding
M1.04	Explain wafer preparation	3	Understanding
Contents: Crystal growth, wafer preparation: Introduction & historical perspective- Clean room concept- Growth of single crystal Si-Czochralski(CZ) and Float zone(FZ) processes. Surface Contamination-chemical-mechanical polishing. Wafer preparation-DI water, Chemical cleaning-RCA, processing consideration- getting the thermal stress factor etc.			
CO2	Explain the process of Epitaxy and Oxidation.		
M2.01	Explain the process of Epitaxy	3	Understanding
M2.02	Illustrate Epitaxial valuation and growth mechanism	3	Understanding
M2.03	Outline the process of oxidation	3	Understanding
M2.04	Describe the kinetics of oxidation	2	Understanding
	Series Test-1	1	

Contents:

Epitaxy and Oxidation: Epitaxy Process-Vapor phase epitaxy, Molecular Beam Epitaxy. Epitaxial Valuation-Growth mechanism and kinetics.

Oxidation techniques-Dry and wet oxidation. Device isolation-LOCOS-Kinetics of Oxidation.

Deal Grove's Model-Oxidation system.

CO3	Explain Lithography and Etching process.		
M3.01	Illustrate the concept of Lithography	2	Understanding
M3.02	Explain different types of Lithographic processes	4	Understanding
M3.03	Explain the concept of etching	1	Understanding
M3.04	Explain the different types of etching process	3	Understanding

Contents:

Lithography & Etching:

Lithography-Overview of Lithography-Radiation Sources-Masks-Photoresist-Components of Photoresist- Optical Aligners-Resolution-Depth of focus-Advanced Lithography-Electron beam Lithography-X-ray Lithography-Ion Lithography.

Etching-Anisotropy-Selectivity-Wet Etching, Plasma Etching, Plasma Properties-Reactive - Ion Etching.

CO4	Illustrate the process of Deposition, Diffusion, Ion implantation and Packaging.		
M4.01	Explain Diffusion mechanism	3	Understanding
M4.02	Explain various ion implantation techniques	3	Understanding
M4.03	Explain different deposition mechanisms	3	Understanding
M4.04	Explain metallization and packaging	3	Understanding
	Series Test-II	1	

Contents:

Deposition, Diffusion, Ion implantation and Packaging: Model of diffusion in solids-Atomic diffusion mechanism-Measurement technique-Range theory-Implantation equipment-Annealing shallow junction-High energy implantation-Deposition process-

Chemical vapor deposition-Physical vapor deposition-Patterning-Metallization and Packaging-Baked IC design.

Text / Reference:

T/R	Book Title/Author
T1	S.M Sze, “VLSI Technology”, McGraw Hill Publication, 2003.
T2	S.K Gandhi, “VLSI Fabrication Principle”, Willy-India Pvt Ltd, 2008.
T3	Suresh Babu “Solid State Device & Technology”, Sanguine, 2005.
R1	J.D Plummer, M.D Deal and Peter B. Griffin, “Silicon VLSI Technology: Fundamentals, Practice and Modelling”, Pearson Education Publication, 2009.
R2	Stephen A. Campbell, “Fabrication Engineering at the Micro and Nano scale”, Oxford University Press, 2013

Online resources:

Sl. No.	Website Link
1.	http://www.asic-world.com
2.	https://edu.ieee.org/eg-aiet/ic-fabrication-ic-tech
3.	https://www.testbench.in